

Raspberry Pi Pico W

QUICK REFERENCE

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Getting Started Guide

KEY SPECIFICATIONS

Microcontroller	RP2040 — dual-core Arm Cortex-M0+ @ 133 MHz
Memory	264 KB SRAM · 2 MB onboard flash
Wireless	2.4 GHz Wi-Fi 802.11n · Bluetooth 5.2 (CYW43439)
GPIO	26 exposed pins · 3 ADC · I2C · SPI · UART · 16 PWM
Power	1.8–5.5 V via VSYS · micro-USB 5 V
Onboard LED	Wired to wireless chip — addressed as "LED"

FIRST-RUN WORKFLOW

1. Flash	Hold BOOTSEL, plug in USB, drop the .uf2 firmware onto RPI-RP2
2. Connect	Open Thonny, select the MicroPython (Pico) interpreter
3. Blink	<code>Pin("LED", Pin.OUT)</code> — toggle to confirm the board runs code
4. Network	<code>network.WLAN(STA_IF)</code> — connect and read <code>ifconfig()[0]</code>

COMMON PINS

3V3 (pin 36)	3.3 V regulated output for sensors
VSYS (pin 39)	Main system input, 1.8–5.5 V
GND	Multiple ground pins along both edges
GP0 / GP1	Default UART0 TX / RX
GP26–GP28	ADC-capable analog inputs

ESSENTIAL COMMANDS (MicroPython REPL)

<code>import machine</code>	Core hardware module
<code>machine.Pin("LED", machine.Pin.OUT)</code>	Onboard LED object
<code>help("modules")</code>	List available modules
<code>machine.freq()</code>	Read current CPU frequency